MILITARY SPECIFICATION

PRINTER, CONTACT, PHOTOGRAPHIC EN-100( )

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of photographic contact printer designated as Printer, Contact, Photographic EN-100( )(see 6.5).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MIL-P-116	- Preservation, Methods Of.
MIL-P-4672	- Paper, Photographic, Black And White, Sensitized.
MIL-D-4827	- Developer, Photographic, Paper, Armed Forces Developer No. 25.
MIL-P-13126B	- Part, Materials, And Processes Used In Electronic Communications Equipment.
MIL-M-13231	- Marking Of Electronic Items.
MIL-F-14072	- Finishes For Ground Electronic Equipment.
MIL-P-55159	- Processing Machine, Photographic Film And Paper EH-29( ).
MIL-D-55162B	- Darkroom, Photographic, Portable ES-38( ).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-252	- Wired Equipment Classification Of Visual And Mechanical Defects.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission And Susceptibility Requirements For the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-810	- Environmental Test Methods.

2.1.2 Other Government documents. The following Government document form a part of the specification to the extent specified herein.

SPI 1G00491	- Special Packaging Instructions.

Copies of specifications, standards, and drawings required by manufacturerers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Printer is a semi-automatic step-and-repeat contact printer for printing 9 by 9 inch (228.6 by 228.6mm) or 9 by 18 inch (228.6 by 457.2mm) aerial camera negative frame size in rolls up to 200 feet (60.96m) long. The negatives are exposed on 9.375 inch (238mm) wide by 200 feet (69.96mm) rolls of photographic paper. Manual dodging is capable of being performed. Masks are provided to permit printing 9 by 9 inch (228.6 by 228.6mm) and 9 by 18 inch (228.6 by 457.2mm) negative sizes.

3.2 First article. When specified (see 6.2b), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.3.1 Cleaning. After assembly, the printer shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material.

3.3.2 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

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3.4 Construction.

3.4.1 Tropicalization of materials. Fungus-inert materials shall be used where practicable. Such materials are listed in Requirement 4 of MIL-STD-454. Use of materials which are nutrients for fungi shall be subject to approval by the contracting officer; the nutrient materials shall be cleaned in accordance with Requirement 9 of MIL-STD-454 and treated as follows:

3.4.1.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated materials will be held to the practical minimum.

3.4.1.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the printer, or during use of the finished items when used for the purpose intended.

3.4.1.3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified service conditions.

3.4.1.4 Statement of treatment. When specified (see 6.2c), a statement of treatment shall describe in detail the materials to be treated and the treating materials and processes that the contractor proposed to use.

3.4.2 Fungus. The printer and assemblies shall show no evidence of viable fungus or corrosion (see 6.4.1) when subjected to the 28-day fungus test as specified in 4.6.3.

3.4.3 Electrical system. The printer shall be capable of operating from 115 volts ±10 percent, 47.5 to 63 Hertz (Hz) power source.

3.4.3.1 Breakdown voltage, insulation. Electrical insulation shall have a minimum breakdown voltage of 1000 volts plus 2 times the operating voltage. After application of the voltage for 10 seconds, there shall be no visible deterioration.

3.4.3.2 Current leakage. When tested as specified in 4.6.4.2, the current leakage shall not exceed 2 milliamperes.

3.4.4 Pneumatic rubber bag platen. The bag shall be black, smooth, uniform, and entirely free from imperfections. The contact surface shall have a plate finish and shall be free from tackiness. The rubber shall contain approximately 55 percent crude stock rubber by volume; shall be approximately 0.065 inch (1.7mm) thick; shall have an original tensile strength of not less than 1800 lbs/sq. in (12.4kPa); and shall have a Shore durometer hardness of approximately 50. The original property of elongation and recovery shall be a minimum of 500 percent elongation with complete recovery. The bag shall have in its upper surface a Schrader, or equal, type valve stem with cap or a type that can be used to remove the valve core. The attachment of the bag shall permit easy removal for replacement of bags.

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3.4.5 Paper carriage. The paper carriage shall accommodate 9.375 inch by 200 feet (238mm by 60.96m) roll paper.

3.4.6 Paper feed mechanism. The paper feed mechanism shall operate automatically when the platen raises to the "up" position. It shall meter any preset length from 1 inch (25.4mm) to 20 inches (508mm) with an accuracy of ±0.25 inch (±6.35mm). Indicating scale shall be calibrated to indicate the length of paper that is being metered.

3.4.7 Cassettes. The printer shall be provided with two separate light-tight cassettes which shall be capable of functioning either as supply or take-up. The cassettes shall permit attachment directly to a processing machine conforming to MIL-P-55159.

3.4.8 Printing glasses. Printing glasses shall be provided which are masked for 9 by 18 inch (228.6 by 457.2mm) prints and 9 by 9 inch (228.6 by 228.6mm) prints. The printing glasses shall be masked with an opaque ceramic paint which is fused to the glass. Provisions for printing the fiducial masks shall be provided by four 0.5 inch (12.7mm) semi-circular openings located midway on each edge of the mask. The printing glass shall be a clear commerical selected grade, free from striae, scratches and any other imperfections which would affect the contact print. The upper edges shall be ground and beveled to a smooth finish so that the edges of the glass cannot scratch the film.

3.4.9 Variable contrast filter assembly. The variable contrast filter assembly shall consist of a roll filter, guide rollers, and filter transport mechanism.

3.4.9.1 Roll filter. The filter shall be a roll of Dupont Mylar or Chronar, or equal, plastic material with Dupont Varigam filters 1, 3, 5, 7, and 10, or equal, dyed into it plus a clear area on each end. Each filter or clear area shall be of sufficient size to cover the entire printing aperture of the printer. No white light shall leak around the edges. The filters shall have characteristics in respect to speed and contrast equal to the respective Dupont filters. The roll filter shall be carried in a moisture-proof bag during initial shipment and storage.

3.4.9.2 Guide rollers. The filter assembly shall have a series of rollers to properly guide and roll filter from the supply roller, under the printing aperture, and to the take-up roller. These rollers shall protect the filter from tearing and scratching. The filter shall be placed in a position where the heat from the printing lamp will not harm it after long usage and yet at a place where the light is sufficiently diffused so that any minor defects or dirt on the filter will not adversely affect the print.

3.4.9.3 Takeup rollers. The roll filter shall be attached to the takeup rollers in such a manner that (a) it will not be inadvertently pulled loose during use and (b) it may be removed when the printer is not in operation.

3.4.9.4 Filter transport mechanism. The filters shall be transported into the printing aperture by means of turning cranks. A filter position indicator shall be provided to index the filter in the printing aperture.

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3.4.10 Counter. An electronic counter shall be provided to record the number of prints up to 99 prints from a single negative. The counter shall stop the printing cycle at the predetermined setting and recycle to zero.

3.4.11 Timer. The timer shall be an integral part of the printer and shall automatically control the transport mechanism, exposure and platen mechanism. The timer shall have an oeprating range from 0.1 to 39.9 seconds.

3.4.11.1 Accuracy. When tested as specified in 4.6.6, all readings shall be within ±5 percent of the mean value obtained.

3.4.11.2 Repeatability. The timer shall have a repeatability accuracy of ±2 percent over the entire operating range of voltage and frequency (see 3.4.3) for any setting of the time interval controls.

3.5 Durability. Whe tested as specified in 4.6.8, the printer shall operate for not less than 100,000 cycles without failure.

3.6 Performance characteristics.

3.6.1 Modes of operation. The printer shall be capable of operating in two modes; step-and-repeat mode, and manual.

3.6.1.1 Step-and-repeat mode. During the printing cycle, exposure shall automatically begin when the platen is in the proper position to maintain the print material in contact with the negative material. At the end of exposure, the platen shall automatically release and the print material shall be advanced the required amount automatically.

3.6.1.1.1. Printing cycle. The printing cycle shall consist of platen closure, print exposure, platen release and paper transport. The entire printing cycle shall be completely automatic and the time consumed for a single cycle shall be not more than three seconds.

3.6.1.2 Manual mode. In this mode of operation, the operator shall be able to manually open and close the platen and to start the exposure by depressing the manual button on the control panel. Cut sheets of positive material shall be utilized in this mode.

3.6.2 Print quality. Sensitized materials exposed in the printing unit shall be free from scratches, abrasions or other marks that affect the quality of the finished image.

3.6.3 Paper transport. The feed path shall be from left to right facing the front of the printer. The print material (paper) shall be automatically advanced. A constant tension on the print material shall be maintained regardless of the speed or the amount of material remaining on the feed or take-up spools.

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3.6.4 Dodging. When tested as specified in 4.6.9.4, the density of the square shall be within the following limits:

a. Set 1. The reading of any one square shall not vary more than +30 percent from the average of all squares. This average density shall be considered the control density for the following determinations.

b. Set 2. The 16 squares forming a larger square that were most affected by the "out" lamp in these prints shall each be further divided into four 0.5 inch (12.7mm) squares. The average of the center four 0.5 inch (1.7mm) squares shall be not more than 85 percent of the control density. The average of squares adjacent to the inner square shall be not more than 25 percent less than the contol density. The average of the next outer ring of squares shall be not more than 20 percent less than the control density.

c. Set 3. The average density of the 16 squres forming a larger square most affected by the "out" lamp shall be not more than 40 percent of the control average density. The average density of the ring two squares wide, outside the 16 squares above shall be not more than 50 percent of the control average density.

d. Set 4.The average of the 16 squares in the center of the exposed portion of this print shall be not less than 20 percent of the control average density. The ring of squares outside the center ring shall be not more than 16 percent of the control average.

3.6.5 Temperature rise. The temperature rise shall be not more than 20 Fahrenheit (11 Celsius) degrees above the ambient temperature at the negative and 30 Fahrenheit (17 Celsius) degrees above the ambient temperature at the variable contrast filter.

3.6.6 Daylight operation. The paper shall not be fogged when the printer is operating under daylight conditions.

3.7 Environmental.

3.7.1.1 Low. During testing as specified in 4.6.11.1.1 the equipment shall meet the following requirements.

Paragraph	Measurement	Performance
3.6	Performance	Full specification requirements

After testing, the equipment shall meet full specification performance for these measurements.

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3.7.1.2 High. During testing as specified in 4.6.11.1.2 the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.6	Performance	Full specification requirements

After testing, the equipment shall meet full specification performance for these measurements.

3.7.2 Humidity. The Printer shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the requirements of 3.6 during the operating times specified in 4.6.11.2.

3.7.3 Altitude (elevation). The equipment shall meet full specification performance for the following measurements both during and after testing as specified in 4.6.11.3.

Paragraph	Measurement
3.6	Performance

3.7.4 Vibration. When tested as specified in 4.6.11.4, the amplitude of vibration of any part, subassembly, or structural member of the printer shall be not more than twice the amplitude of the vibration applied to the printer.

3.8 Electromagnetic interference. The printer shall meet the electromagnetic interference requirements specified in MIL-STD-461 for class IIB. If the contractor considers it desirable to modify the equipment in order to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.9 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.10 Finish. The printer shall be finished in accordance with MIL-F-14072. The final paint film on type I surfaces shall be final film E, semigloss, light gray enamel.

3.11 Marking. Marking shall conform to MIL-M-13231; the front panel marking shall be as described in MIL-M-13231, group 1.

3.11.1 Visibility. Whenever practicable, parts shall be mounted so that their identification markings will be readily visible with minimum disassembly of the printer.

3.11.2 Serial numbers. Each printer shall be serial numbered.

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3.12 Workmanship. The printer shall be manufactured and assemblied in accordance with requirement 9 of MIL-STD-454, and the applicable portions of the following paragraphs in MIL-P-11268:

Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Soldering.
Cleaning of units.
General requirements for securing parts.
Controls.
General requirements for printed wiring and boards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3.1 First article units. When required, and as specified in the contract, the contractor shall furnish first article units of the complete printer.

4.3.2 First article inspection. Unless otherwise specified (see 6.2d), this inspection shall be performed by the Government. It shall consist of the inspection specified for supression in 4.6.12, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B and group C (See tables I, II and III, respectively). After completion of group C tests, conforming units shall be reinspected and shall pass all group A inspections. Failure of any inspection shall be cause for rejection of the first article.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements:

Item	Where required
Finish	3.10
Marking	3.11

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4.5 Quality conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in table I. Discrete lots shall be formed from units that pass inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105, using the general inspection levels, and acceptable quality levels (AQL) specified in table I.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective) Major	Minor
Visual and mechanical	3.12	4.6.14	1.0	4.0
Performance				
Contact test	3.6.2	4.6.9.3	1.0	
Dodging test	3.6.4	4.6.9.4	1.0	
Daylight operation	3.6.6	4.6.9.1	1.0	
Operation	3.4.5; 3.4.6; 3.4.10; 3.6.1	4.6.9	1.0	1/
Scratch	3.6.2	4.6.9.2	1.0	
Timer accuracy	3.4.11.1	4.6.6	1.0	
Timer repeatability	3.4.11.2	4.6.7	for the group	

1/ All defects are major.

4.5.2.1 Order of inspection within Group A. Group A inspection shall be performed in any order satisfactory to the Government.

4.5.3 Group B inspection. Group B inspection shall be as specified in table II. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)
Dielectric	3.4.3.1	4.6.4.1	6.5
Roll filter	3.4.9.1	4.6.5	4.0
Interchangeability	3.9	4.6.13	6.5

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4.5.3.1 Group B sampling. Sampling for group B shall be in accordance with MIL-STD-105, inspection level S-4. AQL shall be as specified in table II.

4.5.3.2 Order of inspection within group B. Group B inspection shall be performed in any order satisfactory to the Government.

4.5.4 Group C inspection. Group C inspection shall be as specified in table III and shall be performed on sample units that have been subjected to and passed groups A and B inspections.

TABLE III. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Order of inspection
Fungus	3.4.2	4.6.3	1
Current leakage	3.4.3.2	4.6.4.2	2
Durability	3.5	4.6.8	3
Temperature rise	3.6.5	4.6.10	4
Temperature	3.7.1	4.6.11.1	5
Moisture resistance	3.7.2	4.6.11.2	6
Elevation	3.7.3	4.6.11.3	7
Vibration	3.7.4	4.6.11.4	8

4.5.4.1 Sampling for Group C inspection. One sample unit shall be selected at random from every 50 units, or fraction thereof, produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.4.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in table III.

4.5.4.3 Group C failure. Action required relative to group C failures shall be as specified in the contract (see 6.2e).

4.6 Methods of inspection.

4.6.1 Test conditions.

4.6.1.1 Temperature and humidity. Unless otherwise specified herein, the tests shall be conducted at an ambient temperature of 77°F ±9F (25°C ±5°C) with a relative humidity of 35 to 70 percent insofar as practicable. When measurements are made under conditions other than specified, they shall be corrected, if necessary, to the foregoing standard conditions.

4.6.1.2 Line voltage. Unless otherwise specified herein, the line voltage during tests shall be 115 ±2 volts, 60 Hz, single phase.

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4.6.2 Test setup and material.

4.6.2.1 Paper for tests. Unless otherwise specified, the paper used for all tests shall conform to MIL-P-4672, type IV, subtype E. The paper shall be air-dried.

4.6.2.2 Developing. The paper shall be developed in Armed Forces Developer No. 25 conforming to MIL-D-2827 diluted 1:2 for 1.5 minutes at 68°F (20°C). Equivalent times shall be used if the developer solution is used at a different temperature.

4.6.2.3 Densitometer. The densitometer used to determine the transmission density of the neutral filter and the reflection density of the print from the filter shall be the Ansco-MacBeth type or equal. The instrument shall read zero for a transmission density when a reading is taken with no film in the machine, or the reading taken under these conditions shall be subtracted from or added to the reading taken when there is film in the instrument. The instrument shall read zero when a piece of photographic paper, not developed but fixed out, is placed in the instrument and a reflection density reading taken. If the instrument does not read zero, the reading taken shall be subtracted from or added to those taken on the prints. These zero readings shall be taken before each group of readings on the film or the paper are taken and used only for the readings taken immediately after the zero readings are recorded.

4.6.2.4 Basic exposure time. The basic exposure time shall be that time required to expose the specified paper, developed as specified to an average density of 0.70 ±0.05 using the neutral density filter instead of a negative. The contrast control shall be set at a number 5 filter.

4.6.2.4.1 Neutral density filter. The neutral density filter shall be Wratten 96 neutral density filter, or equal, having a density of 1.00 ±0.05 and shall be not less than 9 by 9 inch (228.6 by 228.6mm). The filter shall be centered in the printing area when the printer is tested.

4.6.2.5 Halftone screen. A halftone screen, 10 by 20 inches (254 by 508mm) consisting of clear dots or clear squares and black field of film shall be used. The screen shall have from 100 to 133 lines per inch (25.4mm). The screen shall be free from scratches, curl or other imperfections.

4.6.2.6 Step tablets. The sensitometric step tablets used for tests shall increase in increments of square root of two steps.

4.6.2.7 Test film for temperature rise. The test film shall be a Wratten 96 neutral density filter, or equal, with a density of 2.0 or greater, or a negative with a minimum average density of 2.0. The test film shall be 10 by 20 inch (254 by 508mm) in size.

4.6.2.8 Variable contrast sheet filters. The variable contrast sheet filters used in the tests shall be DuPont Varigam filters or equal. They shall be 9 by 9 inch (228.6 by 228.6mm) or larger in size.

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4.6.3 Fungus. The printer and assemblies shall be subjected to the 28-day fungus test specified in MIL-STD-810, method 508.1. For gasket or other non-hermetically sealed equipments, the parts and materials of the printer shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in this open condition. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.4 Electrical.

4.6.4.1 Dielectric. The insulation strength of electrical circuits in the printer shall be tested utilizing the voltage specified in 3.4.3.1. The test potential shall be direct current or peak alternating current of the same value and shall be applied between all wiring and ground. Nonconformance to 3.4.3.1 shall constitute failure of this test.

4.6.4.2 Current leakage. Following subjection to temperature test the leakage shall be determined with the printer connected to an alternating current power source having a potential of 115 volts ±10 percent. This test shall be between a power terminal and ground. Nonconformance to 3.4.3.2 shall constitute failure of this test.

4.6.5 Roll filter. The printer shall be set up with the clear area of the roll filter (see 3.4.9.1) in place. A variable contrast sheet filter (see 4.6.2.8) shall be placed on top of the printing glass. Sensitometric step tablets (see 4.6.2.6) shall be placed in each corner and in the center of the sheet filter. Three prints shall be made with the paper as specified in 4.6.2.1. This procedure shall be performed with Varigam sheet filters No. 1, 3, 5, 6, and 10. The above procedure shall be repeated using the roll filters of the printer rather than the sheet filters. The paper shall cover the full aperture of the printer. The density of the steps in the prints shall be determined using the specified densitometer and recorded for all prints. The density values from the three prints shall be averaged and plotted as shown in MIL-P-4672. The total scale, as defined in MIL-P-4672,of the prints made using the sheet filters shall be within ±0.15 of the prints made using the roll filter in the printer. The log E value taken at a density of 1.00 of the prints made with the sheet filters shall be within ±0.15 of the prints made using the roll filter. Nonconformance to 3.4.9.1 shall constitute failure of this test.

4.6.6 Timer accuracy. The timer shall be tested with an instrument which is accurate to ±0.02 second. Not less than 30 values shall be taken on each setting for not less than 10 settings with the voltage and frequency being different for each of the settings. The settings chosen shall vary over the entire range of the timer. At least 3 of the settings shall be below 0.9 seconds and 7 settings equally divided below and above 5 seconds. The mean time value for each setting shall be determined. Nonconformance to 3.4.11.1 shall constitute failure of this test.

4.6.7 Timer repeatability. The timer shall be tested with an instrument which is accurate to ±0.02 second. A value shall be taken on each setting for not less than 10 settings of the time interval controls over the entire range of voltage and frequency. Repeat the cycle for not less than a total of 10 cycles using the same control setting, voltage, and frequency used during the first cycle. After the last cycle has been completed, the percentage of error for all values taken shall be determined. Nonconformance to 3.4.11.2 shall constitute failure of this test.

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4.6.8 Durability. The printer shall be subjected to a test of 100,000 cycles. After the completion of each 25,000 cycles, the printer shall be loaded with processed negative material having both 9 by 9 inch (228.6 by 228.6mm) and 9 by 18 inch (228.6 by 457.2mm) formats. The positive material shall be 200 feet (60.96m) rolls of the material specified in 4.6.2.1. The paper shall be cycled through the printer in two steps. 100 feet (30.48m) of paper shall be advanced to print from the 9 by 9 inch (228.6 by 228.6mm) format and the remaining 100 feet (30.48m) shall be advanced to print from the 9 by 18 inch (228.6 by 457.2mm) format. Twenty-five prints per selected negative shall be made in the automatic step-and-repeat mode. The paper shall be processed in accordance with 4.6.2.2 and examined for spacing between prints. Nonconformance to 3.5 shall constitute failure of this test.

4.6.9 Operation. The printer shall be set up for operation utilizing the halftone screen (see 4.6.2.5) as a negative and photographic paper specified in 4.6.2.1. Prints shall be made while checking the following:

a. Counter (see 3.4.10).
b. Paper carriage (see 3.4.5).
c. Paper feed mechanism (see 3.4.6).
d. Modes of operation (see 3.6.1).

Nonconformance to 3.6 shall constitute failure of this test.

4.6.9.1 Daylight operation. The printer shall be loaded with a 200 foot (60.96m) roll of paper specified in 4.6.2.1 and placed in operation in automatic mode with printing lamps "off" or removed. While cycling 100 feet (30.48m) of paper through the printer, a 100 watt lighted bulb shall be passed continuously along the outer edges of the paper compartments and transport system within 6 inches (152mm) of the frame. The remaining 100 feet (30.48m) of the paper shall be cycled through the printer under safelight conditions. The processed paper shall be examined for any variation in base fog throughout the 200 foot (60.96m) length of paper. Nonconformance to 3.6.6 shall constitute failure of this test.

4.6.9.2 Scratch test. Prior inspection for scratches and abrasions shall be made on the negative and positive materials before testing. The paper as specified in 4.6.2.1 shall be cycled through the printer in two steps: 100 feet (30.48m) of paper shall be advanced to print from the 9 by 9 inch (228.6 by 228.6mm) format and the remaining 100 feet (30.48m) shall be advanced to print from the 9 by 18 inch (228.6 by 457.2mm) format. 25 prints per selected negative shall be made in the automatic step-and-repeat mode. The paper shall be processed as specified in 4.6.2.2 and examined for continuous scratches, abrasions, and marks which will either damage or affect the quality of the print. A continuous scratch for the purpose of this test shall be defined as a single scratch which extends for more than one inch (25.4mm) on the film or paper, or a scratch which appears at the same point in every picture. An abrasion is defined as furrows or grooves in the emulsion caused by the paper rubbing the surface of the negative material during the paper transport cycle. A continuous scratch or an abrasion will be considered a defect if a visual observation recognizes its existence. Nonconformance to 3.6.2 shall constitute failure of this test.

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4.6.9.3 Contact test. The printer shall be set up with all lamps "on", the No. 10 filter in place, the specified halftone screen (see 4.6.2.5) used as a negative, and photographic paper (see 4.6.2.1) covering the complete printing aperture of the printer. The printing glass and halftone screen should be carefully cleaned prior to printing. Five prints each shall be made using the following exposure times:

a. The basic exposure time is specified in 4.6.2.4 herein.

b. 1.5 times the basic exposure time.

c. 2 times the basic exposure time.

d. 0.5 times the basic exposure time.

The prints shall be developed as specified in 4.6.2.2 and each print shall be examined. Evidence of dark areas on the prints, distortion of the dot pattern of the half-tone screen, dots merging or becoming indistinct on any of the screen prints, or nonconformance to 3.6.2 shall constitute failure of this test.

4.6.9.4 Dodging test. The printer shall be set up for operation and tested as follows:

a. The basic exposure time shall be the time of exposure with all lamps "on" required to produce, with no negative, a density of 0.70 ±0.10 when processed as specified. The paper used shall be type 1, sub-type A, contrast 4, in accordance with MIL-P-4672. It shall be air-dried to a semi-matte surface (see 4.6.2.2).

b. Using the paper and the basic exposure time specified above a set of three 10 by 20 inch (254 by 508mm) prints for each of the following conditions shall be made and processed as specified herein:

(1) Set 1: With all lamps "on."
(2) Set 2: With only the center lamp "out."
(3) Set 3: With only the center lamp and lamps adjacent to it "out."
(4) Set 4: With only center lamp and lamps adjacent to it "on."

c. Each print shall be divided into 200 1-inch (24.4mm) squares and the density of each square, with the exception of squares comprising the outer perimeter, shall be determined with the densitometer (see 4.6.2.3). The outer ring of squares shall be disregarded for all determinations. Nonconformance to 3.6.4 shall constitute failure of this test.

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4.6.10 Temperature rise.

4.6.10.1 Test setup. A thermocouple shall be firmly attached to the center of the test film (see 4.6.2.7) on the emulsion side and to the center of the No. 5 filter on the roll filter. The test film shall be placed in the center of the printing glass with the emulsion side away from the printing lamps and the No. 5 filter shall be rolled into the printing aperture. The temperature rise shall be determined with an instrument accurate to $\pm 2^{\circ}F$ ($\pm 1.1^{\circ}C$).

4.6.10.2 Test method. With an ambient temperature of 115°F (46°C) the printer shall be operated for 1 hour, using 5-second exposures and at least 5 exposures per minute with the test film in place. The temperature rise shall be measured at 15 minute intervals, for a period of one hour. Nonconformance to 3.6.5 shall constitute failure of this test.

4.6.11 Environmental.

4.6.11.1 Temperature test.

4.6.11.1.1 Low temperature. The equipment shall be tested in accordance with Method 502.1, procedure I, of MIL-STD-810. The storage temperature shall be -60°F (-51°C) and the operating temperature shall be 40°F (4.4°C). Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.7.1.1.

4.6.11.1.2 High temperature. The equipment shall be tested in accordance with method 501.1, procedure II, of MIL-STD-810. The operating temperature shall be 115°F (46°C). Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.7.1.2.

4.6.11.2 Humidity. The printer shall be subjected to the humidity test specified in method 507.1, procedure II, of MIL-STD-810, with test measurements during Step 4 and the last 5-hour period of the fifth 48-hour cycle in Step 6. Inability of the printer to meet the requirements of 3.7.2 shall constitute failure of this test.

4.6.11.3 Altitude (elevation). The equipment shall be tested in accordance with method 500.1, procedure I, of MIL-STD-810. Prior to and after testing the equipment shall meet full specification performance for the measurements specified in 3.7.3.

4.6.11.4 Internal vibration test. Internal vibration of the printer shall be measured as follows:

a. Secure the printer directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the printer.

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b. Vibrate the printer successively in three mutually perpendicular directions over a frequency range of 10 to 15 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch (0.76mm).

c. In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

d. Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

e. Nonconformance to 3.7.4 shall constitute failure of this test.

4.6.12 Electromagnetic interference. The printer shall be tested for electromagnetic interference characteristics in accordance with MIL-STD-462. Nonconformance to 3.8 shall constitute failure of this test.

4.6.13 Dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasterners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and mating units, and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.6.14 Visual and mechanical inspection. The printer shall be examined for the defects listed in MIL-STD-252. Nonconformance to 3.12 shall constitute failure of this test.

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4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table V, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE V. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure.
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect or illegible.	Visual

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G00491.

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6. NOTES

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2).
c. Time frame required for the contractor's submission of the statement of treatment of materials (see 3.4.1.4).
d. When first article inspection shall not be performed by the Government (see 4.3.2).
e. Group C failure action (see 4.5.4.3).
f. Level A or B preservation and packing (see section 5).
g. When first article inspection rough handling tests are not required.
h. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The contracting officer should include specific instructions in acquisition documents regarding arrangements for number of samples, examination, test, and approval of first article.

6.4 Corrosion. Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

6.5 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: EN-100W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.6 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER

Preparing activity:
Army - ER
(Project 6740-A145)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-55594A(ER)	PRINTER, CONTACT, PHOTOGRAPHIC EN-100( )

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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